

000228

kaolinite occupies sporadic vugs in the rock and is seen adjoining the limonite in the interstices; also elsewhere brownish cherty material is seen associated with kaolinite in the interstices. No chert grains, ferruginous or clear are seen and no heavy minerals apart from occasional almost fresh unworn zircon grains. Absence of feldspar, ferruginous cherts and near absence of heavy minerals and the angularity of large quartz grains suggest this rock to be a member of the Pandurra Formation. The quartz grain fabric is somewhat similar to specimens A4252, 4254 and 4256 from the Main Open Cut at Mount Gunson.

4. REFERENCES

- 1) Pandurra Thin Sections, I, Memo from PJC to DGT dated 4/9/73.
- 2) Petrography of Mount Gunson Sediments, Pacminex Report PMR 195/72, October, 1972.
- 3) Pettijohn, F.J., 1957: 'Sedimentary Rocks', pub. by Harper & Row, N.Y. and London.
- 4) Williams, G., Turner, F.J., and Gilbert, C.M., 1954 ; 'Petrography', pub. by W.H. Freeman & Co., San Francisco
- 5) Pandurra Thin Sections II, Memo from DGT to PJC dated 23/10/73.

D+P from TNT. Laura. 000229

AMDEh

Doodson - ① No detrital muscovine
② Lithic grains mod. abund.
high % sandy recryst. volcanic.
③ V. few H.M. grains.

000230

Ref. No. PMR. 10/74

Prospect 405

PACMINEX PTY LIMITED

PETROGRAPHIC REPORT

SPECIMENS FROM THE POWERLINE EMBAYMENT AREA

MT. GUNSON, S.A.

P.J. CURTIS

SYDNEY

February, 1974.

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P.C. Thomas }

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R.W. Gersteling

K.J. Maiden

J.M. Heape

P.J. Curtis

1. INTRODUCTION

Specimens A9989 (70749) and A9990 (70750) for examination are from an area close to an E.M. anomaly of Mr. W.J. Langron's, i.e. INPUT anomaly "C". Geological relationships of rocks in the area are largely obscured by sand but as the rocks represented by the specimens above appear to overlie the Pandurra Formation it was thought they may be Cretaceous, derived from the Pandurra Formation (Ref. 1).

2. SUMMARY

A9989 (70749)

This rock has many Whyalla Sandstone characteristics, since much of the material is, or was before silicification similar to the lithology of Whyalla Sandstone (rounded grains, abundant lithic cherts and appreciable heavy mineral content). However there are subangular rock fragments present which are typically Pandurra Quartzite type. Possibly reworked, mixed rock material from Whyalla Sandstone and Pandurra Quartzite sources, therefore later than Whyalla Sandstone, which has since been silicified. Not anything like the carbonate cemented 'Cretaceous Sandstone' previously submitted (Ref. 3).

A9990 (70750)

This rock is a banded sandstone of fine grains and of 'coarse' quartz grains. Unlike the Pandurra Quartzite where roughly banded, there are few large erratics in fine quartz bands. The abundance of 'flattened' grains, both large and small, of opaque material in the interstices and the variety of heavy minerals present suggest the rock to be Corraberra Sandstone. There is however no feldspar present.

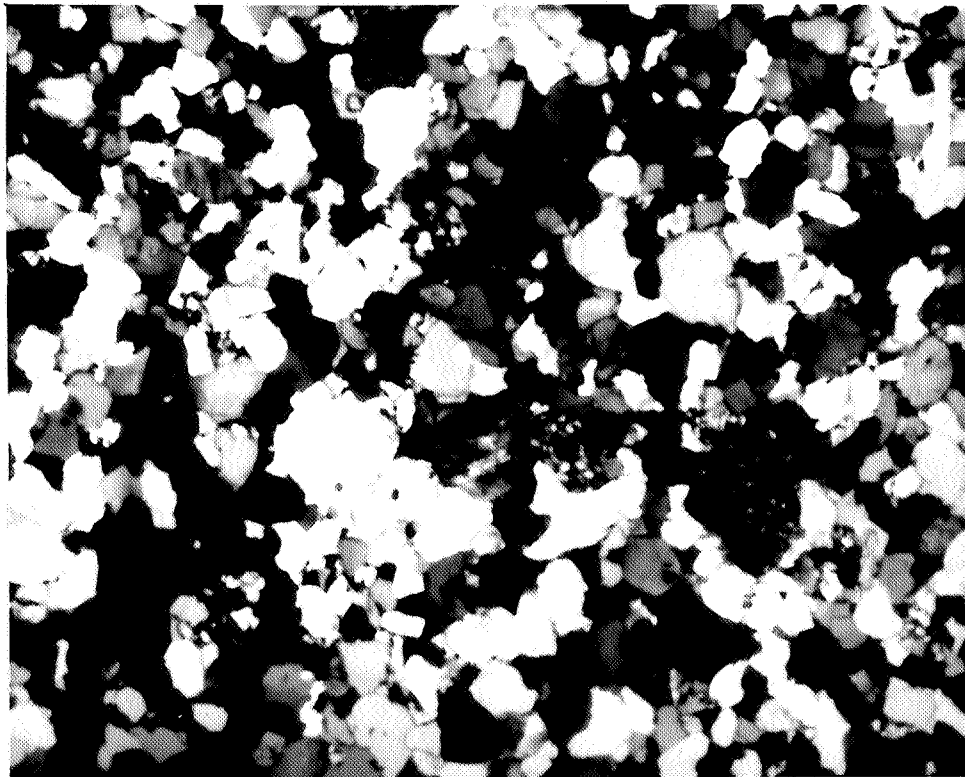


Figure 1: A rock which was formed later than Whyalla Sandstone, from Powerline Embayment area, Mount Gunson.

Note roundness of many quartz grains and angularity of others (flat white, grey or black); and abundance of lithic cherts (speckled black and white) in many parts of photograph.

Crossed nicols

Magnification X 15

3. PETROGRAPHIC INVESTIGATION

A9989 (70749) : (Figure 1)

As seen in hand specimen, an irregular pink quartzite as per Pandurra Formation but with some soft clay matrix. Carbonate occurs as a coating on one surface of the specimen.

Thin section shows many quartz grains, especially the larger grains (approx. 1 mm) to be spherical like those seen in the Whyalla Sandstone. Smaller grains are round, subangular and flat. The range in quartz grains size is from 0.1 to just over 1.0 mm with a median of around 0.4 mm. The grains are very variable in size and are poorly sorted.

The rock is generally well silicified, showing sutured quartz boundaries at contacts between overgrowths of quartz. Many grains, particularly the larger sizes, indicate by trains of opaque inclusions that the primary nuclei were spherical to oval in shape (more spherical than rounded nuclei of Pandurra Quartzite grains which are often made more subangular by secondary silicification); probably rather more than half the grains were in fact quite round before silicification. Fewer than 3 volume percent of quartz grains are rock fragments.

Ferruginous cherts, somewhat rounded, 0.5 to 1.2 mm in size occur from 2 to 3 volume percent of the rock, some show volcanic microlite growths; others are devitrified tachylyte, and about 1 volume percent of clear cherts of similar size to ferruginous cherts are seen.

Kaolinite occurs as small pockets at corners of adjoining quartz grains and in the interstices to just less than 1 volume percent.

A little leucoxene (<1 volume percent) occurs as granular material associated with limonite in occasional vugs or submerged by secondary quartz at the interstices, a little topaz and traces of magnetite as euhedra are also seen.

This is highly silicified rock which is more poorly sorted (as per Pandurra Quartzite) than Whyalla Sandstone generally occurs; but is like Whyalla Sandstone in roundness of many grains, (and of primary quartz nuclei of some silicified grains); abundance of lithic fragments (volcanic chert); and the presence of moderate heavy minerals. There is not quite the roundness of quartz grains as seen in the Whyalla Sandstone of the Plateau area of Mt. Gunson (A4234, A4235) which is also silicified and contains sparse fine matrix (Ref. 2); yet the grains were more rounded before silicification than is seen in Pandurra Quartzite (A4229, Woocalla Quarry) (Ref. 2) to which there are some resemblances in overall fabric; the Pandurra Quartzite in A4241, drill core 3-6-2D 43.6 m, Mystery Area) shows much greater angularity of grains than A9989.

The rock obviously contains much derived Whyalla Sandstone material although a proportion of quartz grains were typically subangular (before silicification) as seen in the Pandurra Formation and the rock is a silicified rock which may be of later age than Whyalla Sandstone.

Comparison with section A4403, Cretaceous Sandstone (?) from Mt. Gunson (Ref. 3) indicates that specimen A9989 is quite different. The carbonate seen on the surface of A9989 was introduced after secondary silicification of the rock, unlike the Cretaceous Sandstone (A4403) which contains abundant carbonate matrix between quartz grains and contains no lithic cherts similar to those seen in A9989.

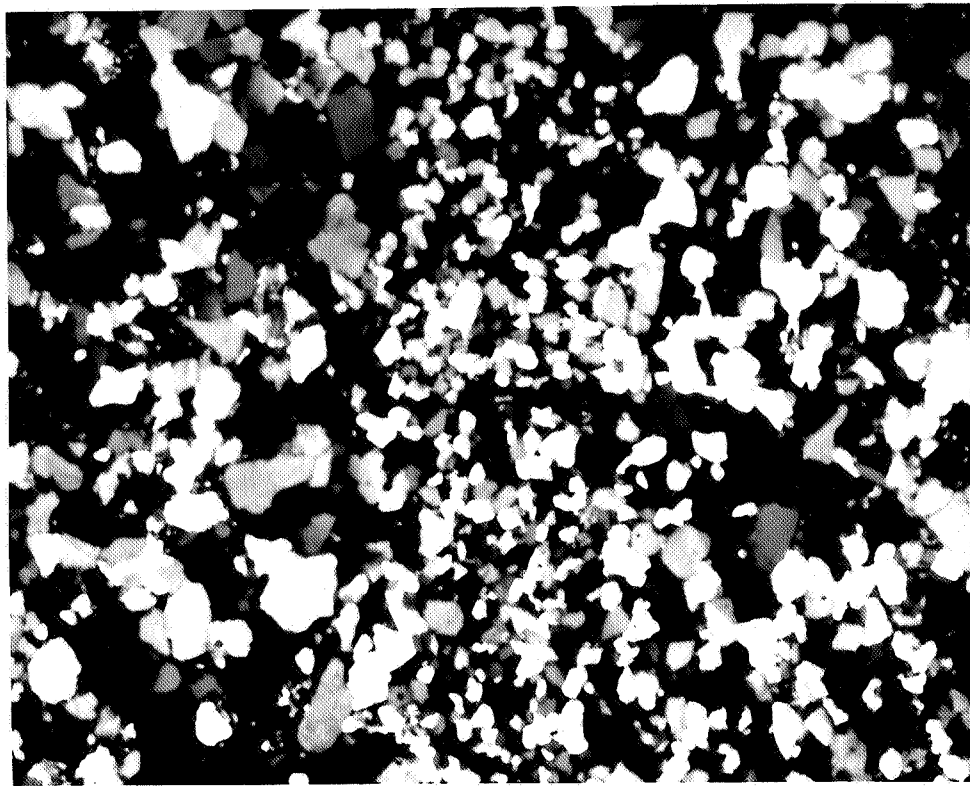


Figure 2 : A rock which is possibly a type of Corraberra Sandstone?

A band of fine quartz grains with few large erratics is seen trending north/south at centre of the photograph. Coarser quartz grains flank the finer band on both sides. Numerous 'flat' grains occur. Careful perusal will show small zircons and sphene grains (white surrounded by black margin, very small), mainly in the central fine quartz area interstices.

Crossed nicols

Magnification X 15

A9990 (70750) : (Figure 2)

A banded sandstone of irregular quartz shapes and sizes. The banding is in fact bedding of coarse and fine quartz grains.

In thin section the quartz grains are seen to be mainly subangular and the different bands vary in thickness up to 3 mm with fine bands of quartz grains which vary between 0.05 to 0.3 mm, median 0.2 mm; and coarser bands varying between 2 and 10 mms, with quartz grains varying between 0.1 and 1.0 mm, median 0.5 mm. In one band of fine quartz occur rare quartz erratics up to 0.7 mm but most fine bands are comparatively free from erratics of much larger grains. Five to 10 percent of quartz grains present in fine and coarse layers are quite flat in appearance. Although about 2 volume percent of quartz grains are somewhat rounded, none is spherical in the Whyalla Sandstone sense.

Only 2 percent of quartz grains are rock fragments and these are of similar size to quartz mineral grains.

Clear cherts are present to about 1 volume percent. No ferruginous cherts are seen.

Much of the interstitial mineral is clear limonite and/or illite flakes which occurs aggregated as a fine film between quartz grains.

Scattered clumps of limonite after magnetite, and leucoxene with limonite (size of aggregates, approx. 0.15 mm) are the most abundant heavy minerals, with less (but still appreciable numbers of) zircons (approx. 0.16 mm in size), sphene and topaz; overall the heavies occur to 2 volume percent. The topaz, zircon and sphene are relatively fresh and all heavy minerals are more concentrated in the finer quartz bands.

Minor silicification of quartz grains is seen.

Probably more like Corraberra Sandstone than any other rock seen in the Mt. Gunson area, and the absence of ferruginous chert and round quartz grains would indicate the paucity of derived Whyalla Sandstone material. Comparing this section with slides from a fairly wide area (Pandurra District and Mt. Gunson area; Refs. 2, 4 and 5) there are resemblances to both Corraberra Sandstone and Pandurra Formation rocks. Similar to Corraberra Sandstone for some quartz grain shapes and variety of heavy minerals and to Pandurra Formation for rough banding with occasional large erratics in otherwise moderately sorted quartz grains. However as previously mentioned, the quartz bands in Pandurra Quartzite ordinarily contain many more 'erratics'.

4. REFERENCES

- 1) Memo to P.J. Curtis from K.J. Maiden dated 22/11/73.
- 2) Pacminex Pty Limited Report PMR. 195/72, entitled, 'Petrography of Mount Gunson Sediments'.
- 3) Memo from P.J. Curtis to K.J. Maiden dated 15/2/73 entitled, 'Mt. Gunson Rocks for Petrographic Examination'.
- 4) Memo to D.G. Tonkin from P.J. Curtis dated 4/9/73 entitled, 'Pandurra Thin Sections'.
- 5) Pacminex Pty Limited Report PMR. 5/74, entitled, 'Petrography of Pandurra Thin Sections II'.

P.J. CURTIS

EMP